

DECODING MUSIC GENRES FROM HUMAN BRAIN ACTIVITY

Whole-Brain fMRI Classification Using Machine Learning

PHASE 1 REPORT

Summary

Objective. This work evaluates whether the genre of a presented 15-second music clip can be decoded from whole-brain functional MRI response patterns. Separate classifiers were developed for five participants and evaluated under strict run-wise separation.

Approach. Each participant contributed 480 unique clip-level observations across 12 training runs and 10 genres. For every cross-validation fold, zero-variance features were removed, the 1,300 highest-scoring voxels were selected using an ANOVA F-test, voxel features were standardized using training data only, and a linear support vector machine was trained. One complete run was held out in every fold.

Outcome. Accuracy ranged from 27.08% to 40.00%, with a mean of 33.92% compared with a 10% dummy baseline. In 1,000 label permutations constrained within runs, no permuted score reached the corresponding observed score for any participant; the empirical p-value was 0.001 for every subject. These results provide strong evidence for statistically reliable within-subject genre decoding from distributed whole-brain fMRI patterns.

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1. Introduction and Research Question

Functional magnetic resonance imaging (fMRI) provides a non-invasive measurement of task-related brain activity through the blood-oxygen-level-dependent (BOLD) signal. Music perception is a distributed cognitive process involving auditory analysis, temporal structure, rhythm, emotion, memory, and motor prediction. Because musical genres differ in combinations of timbre, tempo, rhythm, harmony, and instrumentation, they may evoke distinguishable multivariate response patterns even when no single voxel is sufficient for classification.

The central research question for Phase 1 is:

Can the genre of a music clip be predicted from a participant's whole-brain fMRI response using a leakage-resistant machine-learning pipeline?

The task contains ten genre classes, so a balanced uninformed classifier has a nominal chance level of 10%. The primary goal is not to identify an optimal classifier or a universal cross-participant representation. Instead, it is to establish a statistically defensible within-subject decoding baseline using a fixed model and run-wise cross-validation.

2. Scientific Background

2.1 fMRI and the BOLD response

The BOLD signal reflects local changes in blood oxygenation associated with neural activity. It is indirect, temporally delayed, and affected by physiological and scanner noise.

The dataset used here has a repetition time (TR) of 1.5 seconds; therefore, each 15-second music excerpt corresponds to 10 response volumes.

2.2 Multivariate pattern analysis

Multivariate pattern analysis treats the joint activity pattern across many voxels as one high-dimensional observation. This is appropriate for music perception because genre information may be distributed across several systems rather than isolated in a single region. The analysis is also statistically challenging: each participant has tens of thousands of voxel features but only hundreds of labelled clips.

2.3 Feature selection and support vector machines

An ANOVA F-test ranks each voxel according to the ratio of between-genre to within-genre variability. Retaining the highest-scoring voxels reduces dimensionality while keeping features associated with class differences. A linear support vector machine (SVM) is then used to find maximum-margin decision boundaries in the selected feature space. Linear SVMs are widely used for high-dimensional neuroimaging data and provide a strong, computationally efficient baseline.

2.4 Run-wise validation

Random sample splitting is unsuitable for this dataset because measurements within the same acquisition run share temporal and scanner-specific structure. Leave-One-Run-Out cross-validation holds out all 40 clips from one run and trains on the remaining 11 runs. This tests whether genre information generalizes to a new acquisition run from the same participant.

3. Dataset and Data Integrity

The dataset metadata and event files were obtained from OpenNeuro ds003720 version 1.0.1, while the preprocessed fMRI response matrices were obtained from Zenodo record 8275363. Five participants listened to excerpts from ten genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock.

Property	Value
Participants	5
Training runs per participant	12
Retained clips per run	40
Clips per participant	480
Genres	10
Clips per genre per participant	48
Clip duration	15 seconds
TR	1.5 seconds
Response volumes per clip	10
Nominal chance level	10%

Table 1. Dataset and task summary used in the Phase 1 analysis.

3.1 Subject-specific feature dimensions

Subject	Voxels	Clips	Runs	Genres
sub-001	60,784	480	12	10
sub-002	53,927	480	12	10
sub-003	64,700	480	12	10
sub-004	61,899	480	12	10
sub-005	53,421	480	12	10

Table 2. Clip-level feature dimensions. Voxel counts differ because each participant has an individual brain mask.

3.2 Event audit and clip construction

Each training-run event file contains 41 rows, including an onset-zero stimulus that repeats the final stimulus associated with the preceding run boundary. The analysis excludes this repeated event and retains the remaining 40 unique stimuli in each run.

Retained clip j was mapped to TR indices $10j$ through $10(j+1)-1$, producing a non-overlapping block of exactly 10 response volumes within each run.

Automated assertions verified all of the following before classification:

- exactly 480 observations per participant and 40 observations per run;
- exactly 48 observations for every genre;
- no duplicate retained stimulus identities;
- no duplicate response intervals;
- no response interval crossing a run boundary;

- all 12 runs and all 10 classes present; and
- no non-finite voxel columns in any participant matrix.

4. Methodology

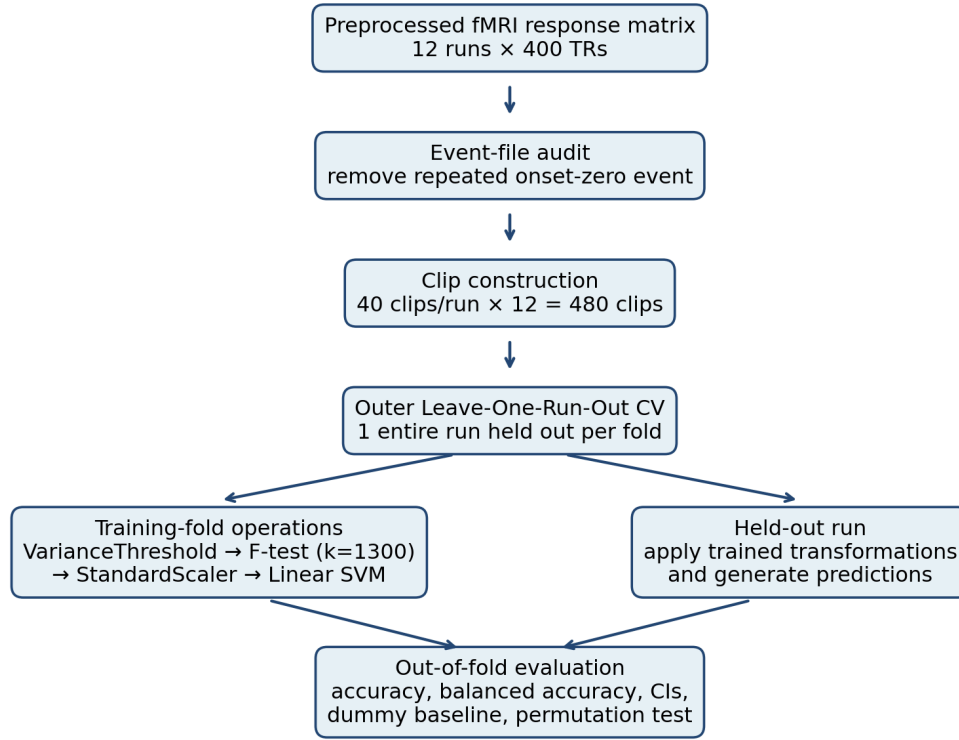


Figure 1. End-to-end Phase 1 workflow. Every learned transformation is fitted using training runs only.

4.1 Clip-level representation

For every retained music excerpt, the 10 response volumes corresponding to its 15-second block are averaged voxel-wise, producing one whole-brain feature vector. This fixed representation is used without subject-specific haemodynamic-delay optimization. The decision avoids selecting temporal parameters from the same cross-validation scores used for evaluation.

4.2 Fold-wise machine-learning pipeline

A single fixed configuration is applied to all five subjects. Within each outer training fold, the following operations are performed in sequence:

- VarianceThreshold removes voxel features with zero variance in the training fold.
- Select K Best with the one-way ANOVA F-test retains the top 1,300 voxels.
- Standard Scaler estimates each selected voxel's mean and standard deviation from the training fold and transforms both training and held-out data using those training statistics.
- A linear SVM is fitted with $C = 1.0$.
- The trained pipeline predicts the 40 clips in the held-out run.

Placing these operations inside one scikit-learn pipeline ensures that the held-out run does not influence feature screening, voxel ranking, scaling, or model fitting.

4.3 Evaluation metrics

Predictions from all 12 held-out runs are concatenated to obtain one out-of-fold prediction for every clip. Overall accuracy and balanced accuracy are reported. Because every genre has exactly 48 observations, the two metrics are numerically identical in this dataset. Accuracy uncertainty was estimated using a 95% confidence interval based on the 12 held-out-run scores.

4.4 Statistical significance

A most-frequent dummy classifier provides a direct baseline. Statistical significance is evaluated separately for each participant using 1,000 permutations. Genre labels are permuted only among clips belonging to the same run, preserving the run structure and class counts while disrupting the relationship between fMRI features and genre. The complete cross-validated pipeline, including voxel selection and scaling, is rerun for every permutation. The empirical p-value is calculated as $(C + 1)/(N + 1)$, where C is the number of permuted scores at least as high as the observed score and $N = 1,000$.

5. Experimental Configuration

Component	Configuration
Feature representation	Mean response across 10 TRs per clip
Primary feature count	$k = 1,300$ voxels
Feature selection	ANOVA F-test, fitted within each training fold
Scaling	StandardScaler, fitted within each training fold
Classifier	Linear support vector machine
SVM regularization	$C = 1.0$
Outer validation	Leave-One-Run-Out, 12 folds
Training/test size per fold	440 / 40 clips
Bootstrap samples	20,000 resamples of run-level scores
Permutation tests	1,000 within-run permutations per participant
Random seed	42
Implementation	Python 3.13.2; NumPy 2.4.4; pandas 3.0.3; scikit-learn 1.8.0

Table 3. Fixed analysis configuration.

6. Results

6.1 Subject-level accuracy

Subject	Accuracy	Balanced accuracy	95% run-bootstrap CI	Chance
sub-001	32.50%	32.50%	28.96%–35.83%	10%
sub-002	34.17%	34.17%	30.21%–38.96%	10%
sub-003	40.00%	40.00%	36.46%–43.54%	10%

Subject	Accuracy	Balanced accuracy	95% run-bootstrap CI	Chance
sub-004	27.08%	27.08%	25.42%–28.54%	10%
sub-005	35.83%	35.83%	32.08%–39.38%	10%
Mean	33.92%	—	—	10%

Table 4. Out-of-fold classification results from the fixed primary pipeline.

The mean accuracy across participants was 33.92%, more than three times the 10% nominal chance level. Every participant exceeded chance, with performance ranging from 27.08% for sub-004 to 40.00% for sub-003.

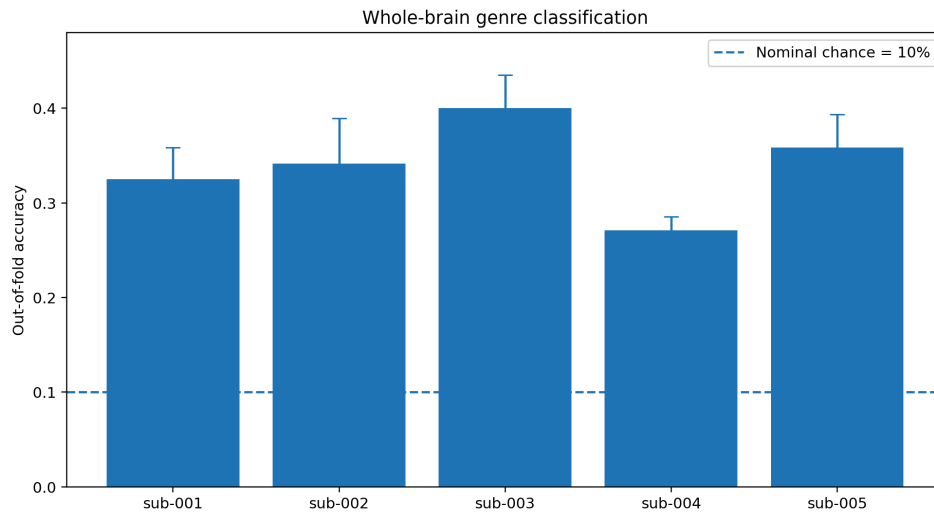


Figure 2. Subject-level out-of-fold accuracy. Error bars show 95% bootstrap intervals over the 12 held-out-run scores.

6.2 Variation across held-out runs

Accuracy varied across runs, as expected for noisy neuroimaging data, but the subject-level curves generally remained well above the 10% reference line. No participant's result depended on a single train-test split because every run served once as the held-out set.

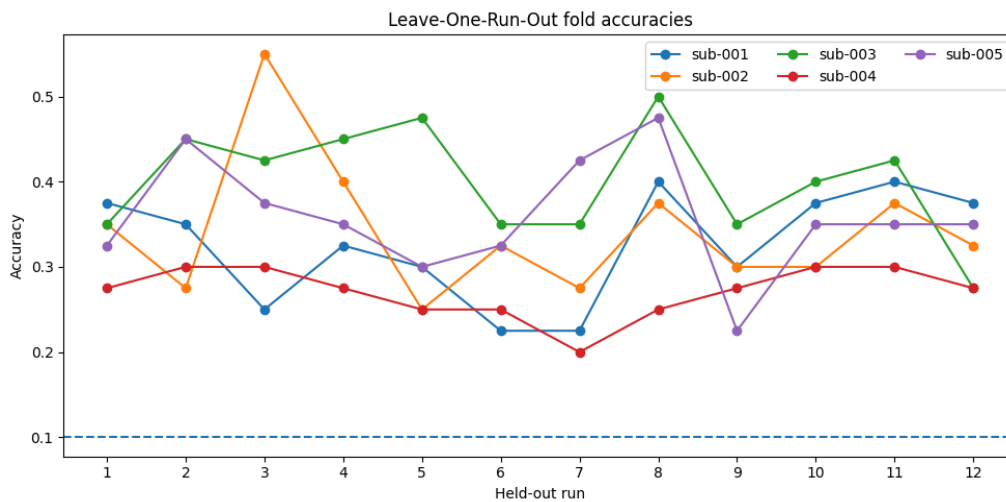


Figure 3. Accuracy for each held-out run and participant.

6.3 Dummy baseline and permutation testing

The most-frequent dummy classifier achieved exactly 10% accuracy and balanced accuracy for every participant. The permutation distributions were centred close to 10%, with standard deviations of approximately 1.4 - 1.5 percentage points. None of the 1,000 permuted scores reached the corresponding observed score for any participant.

Subject	Observed	Permutation mean	Permutation SD	Empirical p	Permutations
sub-001	32.50%	10.00%	1.47%	0.001	1,000
sub-002	34.17%	10.05%	1.44%	0.001	1,000
sub-003	40.00%	9.97%	1.49%	0.001	1,000
sub-004	27.08%	9.89%	1.44%	0.001	1,000
sub-005	35.83%	10.04%	1.55%	0.001	1,000

Table 5. Within-run permutation-test results. The displayed p-value is rounded from 0.000999.

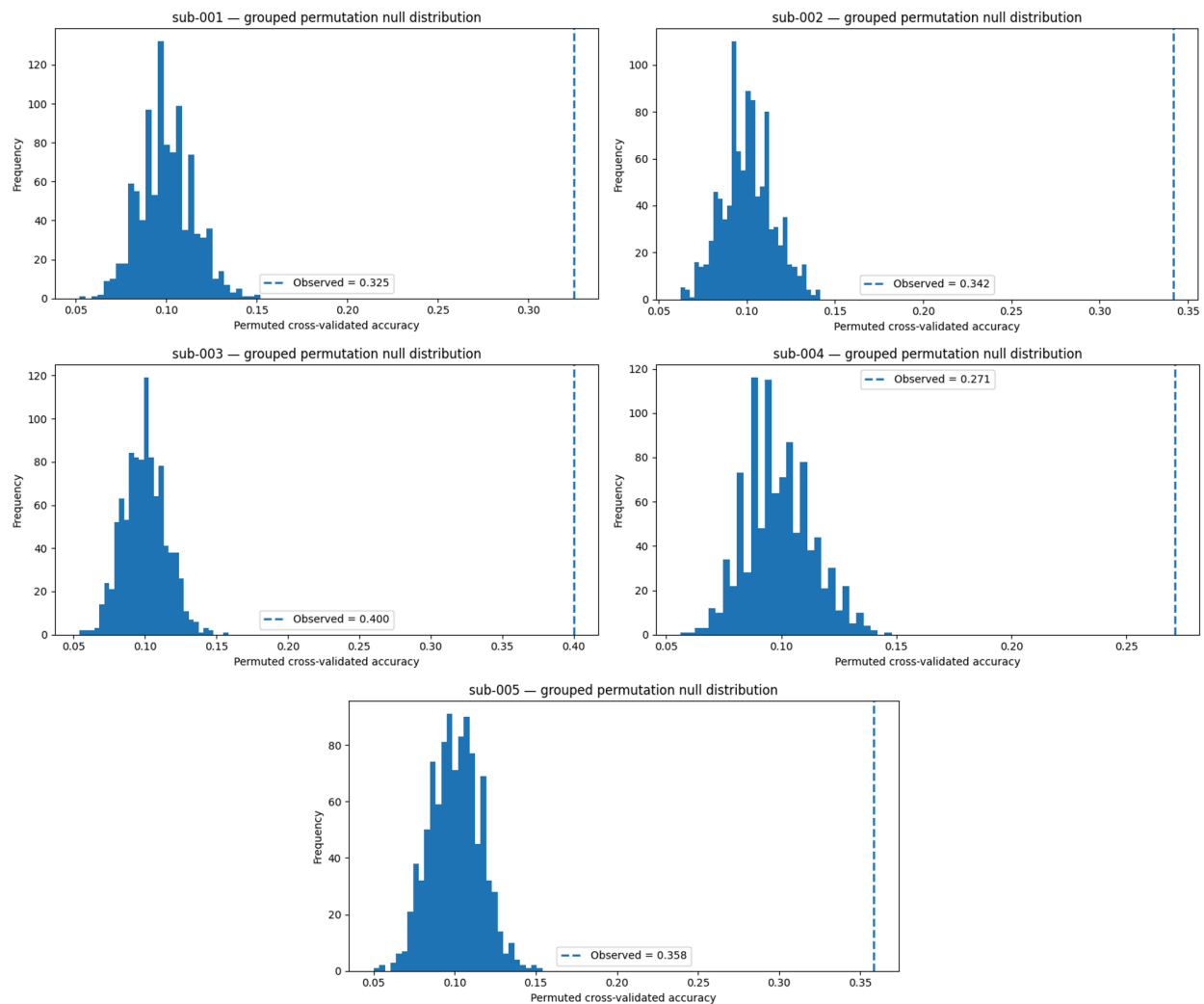


Figure 4. Null distributions from 1,000 within-run label permutations per participant. Dashed lines indicate observed accuracy.

6.4 Confusion patterns and genre-level performance

The confusion matrices show that the classifier did not collapse to one preferred class. Performance differed substantially by genre and participant. When precision, recall, and F1 scores were averaged descriptively across the five independently trained subject models, classical and hip-hop were the most reliably decoded genres, while rock was the least reliably decoded.

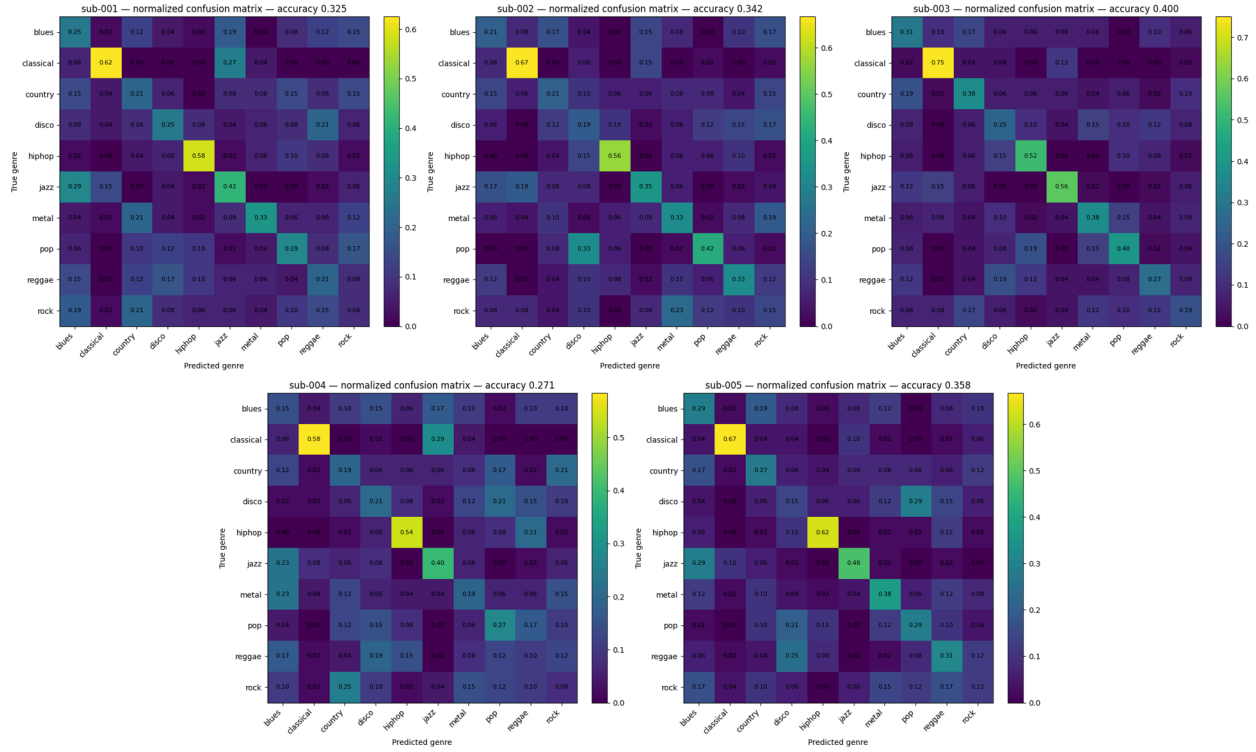


Figure 5. Normalized out-of-fold confusion matrices for all five participants.

Genre	Mean precision	Mean recall	Mean F1
classical	0.670	0.658	0.661
hip-hop	0.565	0.567	0.566
jazz	0.441	0.442	0.440
pop	0.346	0.333	0.339
metal	0.343	0.321	0.330
country	0.247	0.250	0.248
reggae	0.250	0.246	0.247
blues	0.210	0.242	0.224
disco	0.206	0.208	0.206
rock	0.147	0.125	0.135

Table 6. Descriptive per-genre metrics averaged across participants.

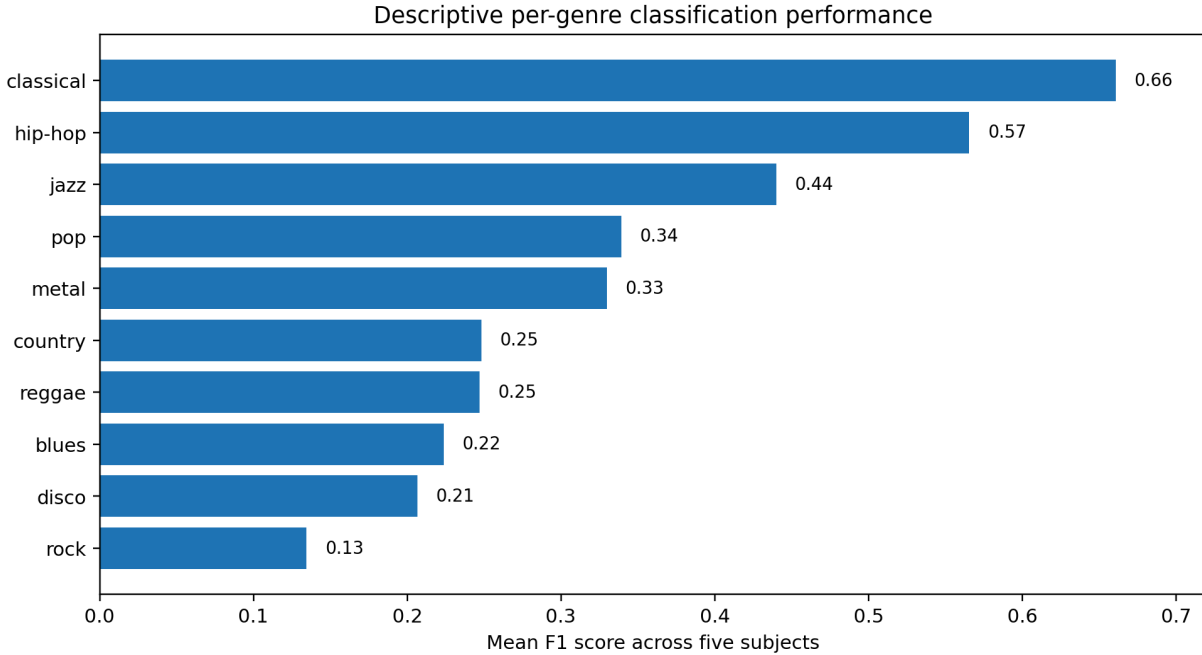


Figure 6. Mean F1 score by genre. These values are exploratory and are not accompanied by genre-wise inferential tests.

7. Discussion

The Phase 1 experiment demonstrates statistically reliable within-subject decoding of ten music genres from whole-brain fMRI response patterns. A mean accuracy of 33.92% on a balanced ten-class task represents substantial information above the 10% baseline. The convergence of three forms of evidence- out-of-fold accuracy, dummy-classifier comparison, and permutation testing supports the conclusion that the model is exploiting a reproducible relationship between distributed BOLD patterns and the genre of the presented stimulus.

Sub-003 achieved the strongest accuracy at 40.00%, while sub-004 achieved 27.08%. Such variation is plausible in participant-specific fMRI analyses because individuals differ in anatomy, signal-to-noise ratio, motion, attention, and the consistency of neural responses. Importantly, all five participants remained statistically significant under the same fixed model and evaluation procedure.

The genre-level pattern is scientifically interesting but should be treated as descriptive. Classical and hip-hop yielded the largest average F1 scores, whereas rock produced the smallest. These differences may reflect acoustic distinctiveness, response consistency, or confusions among neighbouring genres. Anatomical localization and genre-wise inferential testing are required before making claims about specific neural representations.

8. Limitations

- Only five participants were analysed - the results establish consistency across these individuals but do not support broad population-level claims.
- All classifiers are participant-specific. The analysis does not test whether a model trained on some participants generalizes to an unseen participant.
- The official six test runs and their repeated stimuli were not used. Current performance is based on run-wise cross-validation within the training portion of the dataset.

- Temporal averaging compresses each 15-second response block into one vector and discards within-clip response dynamics.
- The fixed values $k = 1,300$ and $C = 1.0$ provide a defensible baseline but are not claimed to be optimal.
- The selected voxels have not yet been mapped to anatomical regions or tested for stability across folds.
- The subject-level permutation tests establish above-chance decoding, but a separate group-level inferential framework would be needed for a formal population-level statement.

9. Future Work

- Evaluate the frozen pipeline on the official held-out test runs, using the repeated presentations to improve response reliability.
- Use nested run-wise cross-validation if kernel, C , gamma, voxel count, or temporal parameters are tuned.
- Register subject-specific feature masks to a common anatomical space and evaluate auditory and music-related regions of interest.
- Compare rectangular clip averaging with canonical-HRF, finite-impulse-response, or run-wise GLM estimates.
- Perform controlled ablation experiments to quantify the contributions of standardization, feature count, and temporal representation.
- Compare linear SVM performance with regularized logistic regression, linear discriminant analysis, and other interpretable baselines.
- Investigate cross-subject alignment methods and leave-one-subject-out generalization.

10. Conclusion

A whole-brain classification pipeline successfully decoded the genre of presented music from fMRI responses in all five participants. The analysis used 480 unique clip-level observations per subject, fold-wise ANOVA voxel selection, fold-wise standardization, a linear SVM, and Leave-One-Run-Out validation. Accuracy ranged from 27.08% to 40.00%, with a mean of 33.92% compared with a 10% baseline. All five results were significant under 1,000 within-run label permutations, with empirical $p = 0.001$. This establishes a strong and reproducible Phase 1 baseline for subsequent held-out testing, anatomical interpretation, and model optimization.

References

- [1] OpenNeuro. *Music Genre Neuroimaging Dataset*, ds003720, version 1.0.1. <https://openneuro.org/datasets/ds003720/versions/1.0.1>
- [2] Nakai, T., Koide-Majima, N., & Nishimoto, S. Preprocessed music-genre fMRI response data. Zenodo record 8275363. <https://doi.org/10.5281/zenodo.8275363>